GUARD Project: Ensuring Interoperable and Trustworthy Knowledge Graphs for Defence and National Security AI*

Jonathon Dilworth[1], Pedro Cotovio[1,2] and Ernesto Jiménez-Ruiz[1,*]

[1]City St George's, University of London, UK
[2]University of Lisbon, Portugal

Abstract

Knowledge graphs (KGs) and ontologies are essential for trustworthy AI in the defence and national security domain, yet their effectiveness hinges on two key properties that are difficult to guarantee at scale: (i) interoperability across independently developed resources, and (ii) the quality of the integrated result. The GUARD project aims to address both in two phases. The first phase, concerned with interoperability and coverage, has produced LogMapLLM, an open-source LLM-enhanced ontology alignment system, and DISO, an openly licensed network of more than sixty defence, intelligence and national security ontologies, accompanied by a reproducible consensus-based pipeline and a new OAEI track. The second phase, which is focused on KG quality, aims to develop scalable logic-based validation and repair techniques for KG integration; we approach this by combining OWL 2 RL and Datalog reasoning together with LLM-assisted correction. This paper presents a consolidated view of the GUARD ecosystem, how its artefacts connect, what is already openly available for community use, and the roadmap ahead. We invite researchers and practitioners to use, evaluate, and extend these resources.

Keywords

Knowledge Graphs, Ontology Alignment, Defence and Security, Large Language Models

1. Introduction

Knowledge graphs (KGs) and ontologies are key enablers of trustworthy AI, but interoperability gaps and quality issues often limit their effectiveness. Upper and mid-level interoperability has been advanced by initiatives such as the Basic Formal Ontology (BFO) [2] and the Common Core Ontologies (CCO) [3]. However, achieving effective application-level interoperability remains a major challenge, particularly in defence and security, where broad conceptual coverage meets highly specialised subdomains. An ontology foundry for this domain, such as that proposed by the US Department of Defence/War, must guarantee interoperability, coverage, and quality—or risk undermining mission-critical downstream applications. The GUARD project aims to address two key challenges:

1. **Interoperability and coverage.** Ontologies and KGs relevant to defence model a broad set of overlapping domains yet interoperate poorly; coverage must be extended through linked-data resources spanning geolocations, organisations, diseases, and environmental hazards.

2. **Quality.** Integration itself is often problematic; alignments can cause merged ontologies to entail contradictions, and instance data drifts out of conformance. Detection and repair at scale is difficult—complete reasoning is often intractable, and its scalable approximations miss violations.

Phase one of GUARD (November 2025 – April 2026) addressed the first challenge through cost-effective use of large language models (LLMs) for KG integration, delivering LogMapLLM [4], an open-source LLM-enhanced ontology alignment system built on LogMap [5], and DISO, an openly licensed network of defence, intelligence and security ontologies [6] with a study of their semantic

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ernesto.jimenez-ruiz@citystgeorges.ac.uk (E. Jiménez-Ruiz)
https://www.citystgeorges.ac.uk/about/people/academics/ernesto-jimenez-ruiz (E. Jiménez-Ruiz)
0000-0002-9083-4599 (E. Jiménez-Ruiz)



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overlap [1]. Phase two (June 2026 – May 2027) addresses the second challenge with improved repair techniques, efficient logic-based KG validation, and LLM-based support for (potential) error correction.

2. The GUARD Ecosystem

The GUARD ecosystem, described in Figure 1 and Table 1, includes resources (the DISO network and the DISO-OAEI track) and systems (LogMapLLM and the DISO-mappings consensus pipeline). Both systems and resources complement one another. DISO provides the collection of domain-clustered ontologies used by matchers such as LogMapLLM, whose participation in the DISO-mappings consensus pipeline yields the reference alignment used in DISO-OAEI. As additional matchers are included under the evaluation initiative, the reference alignment is continually refined. Together, these resources and systems constitute GUARD's answer to the first key challenge under phase one (interoperability and coverage). They also set the stage for phase two: the alignments produced by the ecosystem (and the logical errors that systems introduce) function as the input to the validation and repair framework outlined in Section 4.

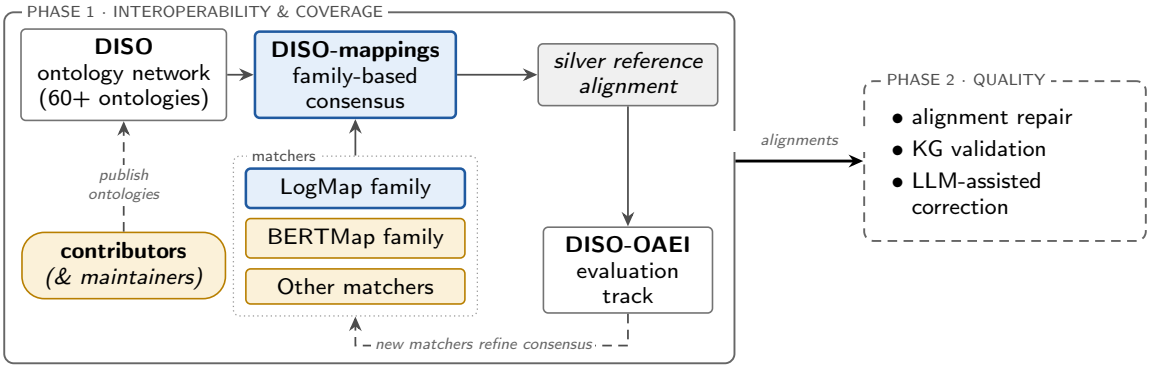


Figure 1: The GUARD ecosystem. Blue boxes are systems (DISO-mappings; the LogMap family, incl. LogMapLLM); white: resources (Table 1); orange: external—third-party matcher families and contributors. Dashed arrows mark ongoing growth: contributors publish ontologies, new matchers participating in DISO-OAEI refine the consensus. Phase 2 (Section 4) consumes the resulting alignments.

Table 1
Openly available GUARD artefacts (Phase 1).

Artefact	Description	Availability
LogMapLLM *(matching system)*	LLM-enhanced ontology alignment system built on LogMap; prompt templates; benchmark for LLM oracle selection [4]	https://github.com/city-artificial-intelligence/logmap-llm
DISO *(network of ontologies)*	Openly licensed network of 60+ public defence, intelligence and security ontologies, clustered by subdomain [6, 1]	https://github.com/city-artificial-intelligence/diso; DOI 10.5281/zenodo.20059506
DISO-mappings *(consensus-based alignment pipeline)*	Reproducible, extensible matching pipeline (AML, LogMap, LogMapLt, BERTMap, BERTMapLt, …) with family-based consensus voting producing a silver-standard reference alignment [1]	https://github.com/city-artificial-intelligence/DISO-mappings
DISO-OAEI *(campaign benchmark)*	OAEI track: global alignment over six ontology pairs with repaired silver references, and local equivalence ranking with NIL mappings; CodaBench evaluation [1]	https://github.com/city-artificial-intelligence/diso-oaei

3. Phase One: Interoperability and Coverage

3.1. LogMapLLM

Ontology matching in specialised high-stakes domains often requires input from experts. Conveniently, tools such as LogMap support interactive matching, in which an external oracle [7] is consulted for correspondences the system cannot confidently accept or reject. LogMapLLM [4] extends this function to route consultation requests to an LLM, allowing it to adjudicate those uncertain candidate correspondences when consulting a human expert is infeasible (due to, e.g., time, cost, and so forth). The system supports both self-hosted, open-weight models[1] for maximal reproducibility, privacy—a practical requirement in the defence and security sector—and cost efficiency, as well as routing to proprietary models for frontier-grade performance.

This work has led to: (i) an open-source release of the LogMapLLM architecture, (ii) a library of domain-specific and domain-agnostic prompt templates for class, property, and instance matching, with retrieval-augmented few-shot examples derived from LogMap's anchors in a semi-supervised fashion, (iii) LogMap-Bio-LLM, a variant that derives composed mappings through mediating ontologies, and (iv) an ongoing benchmarking effort to aid users in LLM oracle selection, with plans to use this benchmarking data for bootstrapping an automatic model-prompt selector.

3.2. DISO: Defence, Intelligence, and Security Ontologies

The **DISO network** is the product of a systematic survey, conducted in late 2025, of the public ontology landscape relevant to defence and national security (as per our inclusion criteria specified below). The result is a curated collection of more than sixty OWL ontologies, each documented with its accompanying metadata, organised into subdomain clusters. The network itself is redistributable, since we sought and obtained explicit permission where required, with the outcome recorded in a per-ontology licensing table. The collection is archived and versioned on Zenodo, with major releases intended to track the annual OAEI cycle, per the released track[2], and a merged "compact" distribution provides one canonical file per ontology for downstream consumers (e.g., our consensus pipeline). The accompanying resource paper [1] analyses the semantic overlap between the network's clusters and discusses the employed reference alignment generation procedure.

Alignments across the DISO network are obtained via **DISO-mappings**, an open-source pipeline that drives a set of configurable matchers. Each matcher is grouped by family, and their outputs are aggregated via *family-consensus-based voting*; this ensures any family with a disproportionate number of matchers cannot inflate agreement. We derive our silver-standard (partial) reference alignment, for use within DISO-OAEI, by accepting correspondences that at least two families agree on.

DISO-OAEI leverages the consensus for use in a new OAEI track comprising two tasks. The first task involves global alignment over six ontology pairs from the command-and-control and infrastructure clusters, with partial reference alignments that were manually reviewed and repaired—removing or weakening correspondences flagged by Alcomo ∪ LogMap ∪ AML [10, 11]. The second task requires participants to perform equivalence ranking over two cyber-security pairs across pooled candidates, with explicit NIL abstention (linking to a special NIL entity [12]). Evaluation runs on CodaBench [13], with planned MELT [14] evaluation to follow.

What makes an ontology a "defence and security" ontology? This is the question we are regularly asked, and one any domain foundry must be able to answer. DISO adopts working inclusion criteria along two dimensions. *Domain fit:* the ontology's core subject matter lies within one or more of the recurring subdomains that defence and security applications typically compose; those being, threats (physical and cyber), actors (state, non-state, and intelligence), operations (military and emergency response), infrastructure and facilities, policy and law, and geospatial context; or the ontology plays

[1] for use in combination with an inference engine, such as vLLM or Sglang [8, 9].
[2] https://diso.oaei-ml.org/

an *enabling role* for such applications, for instance, as smart-building and urban ontologies do for critical-infrastructure protection. *Resource fit for membership in DISO:* the artefact is a publicly available OWL ontology with a licence compatible with redistribution. The interpretation is deliberately broad. Each cluster's inclusion is justified, with sources, in the repository documentation.

4. Phase Two: Quality—Repair & Validation (Future work)

Phase two concerns *KG quality*, and will investigate *alignment repair techniques* and *KG validation*. For any pair of ontologies, $\mathcal{O}_1$ and $\mathcal{O}_2$, an alignment $\mathcal{M}$ is coherent if no named class of $\mathcal{O}_1$ or $\mathcal{O}_2$ is unsatisfiable in the merged ontology $\mathcal{O}_1 \cup \mathcal{O}_2 \cup \mathcal{M}$ [10], a condition that many automatic matchers routinely violate. *Mapping repair* aims to restore coherence by removing a (small) subset of correspondences in $\mathcal{M}$ (or weakening equivalence to directional subsumption), while preserving as much of the original alignment as possible. Our roadmap comprises three directions, all to be exercised on the alignments produced by the ecosystem described by phase one.

Alignment repair beyond Horn-propositional logic. In practice, matchers that perform repair will use *approximate reasoning* to maintain tractability. LogMap, for instance, projects the merged ontology into propositional Horn rules, where propositional variables represent named classes (from subsumption and equivalence axioms, explicit disjointness, and the mappings themselves), and satisfiability is tested using a Dowling–Gallier linear-time Horn-SAT procedure [15]. This projection, however, has a *restriction blind spot*, such that any axiom involving an anonymous class expression (e.g., existential and universal restrictions, cardinalities) is discarded; no property hierarchy, functionality, or inverse axioms are considered, and property mappings are anchored only after class repair has finished, meaning they never enter the Horn clause set at all. Restriction and property-based violations can therefore evade detection entirely. We will explore encodings beyond Horn-propositional logic, notably OWL 2 RL [16], which captures much of this missing vocabulary while staying tractable for Datalog engines proven to scale to large KGs (e.g., RDFox [17]).

OWL axioms as integrity constraints. Quality is also a matter of the data conforming to its ontology, and OWL's open-world semantics make it a poor fit for integrity checking. SHACL [18] and ShEx [19] offer a potential solution, but their constraints must be authored in addition to the ontology, making their scalability a concern (that constraint learning [20] only partially offsets). Building on prior work [21], our approach is to interpret selected OWL 2 axioms as integrity constraints and automatically transform their representation to Datalog rules with their counterpart SHACL or ShEx constraints.

LLMs in the repair loop. Although LLMs have been widely applied to KG completion, the combination of logic-based violation *detection* with LLM-based *correction* has received little attention. We propose the following two approaches. First, when validation identifies a missing fact—for instance, a military asset lacking its required location in the KG—the LLM generates candidate completions that can be checked downstream. Second, when an alignment repair diagnosis yields multiple plans (e.g., removing a mapping versus weakening another), the LLM makes a choice between them. Both approaches may reuse the LogMapLLM architecture already established during phase one.

5. Conclusion

We introduced GUARD, a project investigating interoperability, coverage, and KG quality for trustworthy AI in defence and national security. We presented its ecosystem of open-source artefacts the community can use today, detailed phase one, and outlined phase two. We invite readers to participate in the first edition of the DISO-OAEI track and to propose additional ontologies for the DISO network via the repository issue tracker. Phase two's validation and repair tooling will be released openly as it continues to mature.

Declaration on Generative AI

During the preparation of this work, the authors used Grammarly to check grammar, spelling, and readability; and Google Search with Gemini to suggest alternative phraseology. After using these tools, the authors reviewed and edited the content as needed to take full responsibility for the publication's content.

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References

[1] J. Dilworth, P. G. Cotovio, D. Herron, P. Cripps, N. Dewdney, C. Pesquita, E. Jiménez-Ruiz, Improving Interoperability among Defence and National Security Ontologies, in: The Semantic Web – ISWC 2026, LNCS, Springer, 2026.

[2] J. N. Otte, J. Beverley, A. Ruttenberg, Bfo: Basic formal ontology, Applied ontology 17 (2022) 17–43.

[3] M. Jensen, G. De Colle, S. Kindya, C. More, A. P. Cox, J. Beverley, The common core ontologies, arXiv preprint arXiv:2404.17758 (2024).

[4] S. Lushnei, D. Shumskyi, S. Shykula, E. Jiménez-Ruiz, A. d. Garcez, Large Language Models as Oracles for Ontology Alignment, in: Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), 2026, pp. 2435–2449.

[5] E. Jiménez-Ruiz, B. Cuenca Grau, LogMap: Logic-based and scalable ontology matching, in: International Semantic Web Conference, Springer, 2011, pp. 273–288.

[6] J. Dilworth, P. Cotovio, D. Herron, C. Pesquita, E. Jimenez-Ruiz, P. Cripps, N. Dewdney, DISO: Defence, Intelligence and Security Ontologies , 2026. URL: https://doi.org/10.5281/zenodo.20760510. doi:10.5281/zenodo.20760510.

[7] Z. Dragisic, V. Ivanova, P. Lambrix, D. Faria, E. Jiménez-Ruiz, C. Pesquita, User validation in ontology alignment, in: The Semantic Web – ISWC 2016, volume 9981 of *LNCS*, Springer, 2016, pp. 200–217.

[8] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, I. Stoica, Efficient memory management for large language model serving with pagedattention, in: Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles, 2023.

[9] L. Zheng, L. Yin, Z. Xie, C. Sun, J. Huang, C. H. Yu, S. Cao, C. Kozyrakis, I. Stoica, J. E. Gonzalez, et al., Sglang: Efficient execution of structured language model programs, Advances in neural information processing systems 37 (2024) 62557–62583.

[10] C. Meilicke, Alignment Incoherence in Ontology Matching, Ph.D. thesis, University of Mannheim, 2011.

[11] D. Faria, C. Pesquita, E. Santos, M. Palmonari, I. F. Cruz, F. M. Couto, The AgreementMakerLight ontology matching system, in: OTM Confederated International Conferences" On the Move to Meaningful Internet Systems", Springer, 2013, pp. 527–541.

[12] H. Dong, J. Chen, Y. He, Y. Liu, I. Horrocks, Reveal the unknown: Out-of-knowledge-base mention discovery with entity linking, in: Proceedings of the 32nd ACM international conference on information and knowledge management, 2023, pp. 452–462.

[13] Z. Xu, S. Escalera, A. Pavão, M. Richard, W.-W. Tu, Q. Yao, H. Zhao, I. Guyon, Codabench: Flexible, easy-to-use, and reproducible meta-benchmark platform, Patterns 3 (2022).

[14] S. Hertling, J. Portisch, H. Paulheim, MELT - matching evaluation toolkit, in: International conference on semantic systems (SEMANTICS), 2019, pp. 231–245.
[15] W. F. Dowling, J. H. Gallier, Linear-time algorithms for testing the satisfiability of propositional Horn formulae., J. Log. Prog. 1 (1984) 267–284.
[16] M. Krötzsch, Owl 2 profiles: An introduction to lightweight ontology languages, in: Reasoning Web International Summer School, Springer, 2012, pp. 112–183.
[17] Y. Nenov, R. Piro, B. Motik, I. Horrocks, Z. Wu, J. Banerjee, RDFox: A Highly-Scalable RDF Store, in: International Semantic Web Conference, 2015, pp. 3–20.
[18] H. Knublauch, D. Kontokostas, Shapes constraint language (shacl), W3C Candidate Recommendation 11 (2017) 1.
[19] E. Prud'hommeaux, J. E. Labra Gayo, H. Solbrig, Shape expressions: an rdf validation and transformation language, in: Proceedings of the 10th International Conference on Semantic Systems, 2014, pp. 32–40.
[20] P. G. Omran, K. Taylor, S. J. R. Méndez, A. Haller, Learning shacl shapes from knowledge graphs., Semantic Web 14 (2023) 101–121.
[21] E. Kharlamov, B. C. Grau, E. Jiménez-Ruiz, S. Lamparter, G. Mehdi, M. Ringsquandl, Y. Nenov, S. Grimm, M. Roshchin, I. Horrocks, Capturing industrial information models with ontologies and constraints, in: International Semantic Web Conference, Springer, 2016, pp. 325–343.